Baseline Retrieval vs. Cross-Encoder Re-Ranking

Github link: https://github.com/Rajarshi-1998-Chatterjee/2025814004_LMA_A1

Setup (from `results/run_config.json`): scanned 22-page PDF → RapidOCR PP-OCRv6 (medium) → 307 chunks (≤450 chars, 1-sentence overlap). Bi-Encoder `all-MiniLM-L6-v2` + FAISS inner product (Top-10); Cross-Encoder `ms-marco-MiniLM-L-6-v2` re-sorts those 10; Top-3 of each pipeline go to `Qwen2.5-7B-Instruct` (bf16, greedy, identical prompt). GPU: A100-40GB. 9 test queries (7 failure-type, 2 controls).

1. Architecture comparison

A **Bi-Encoder** encodes the query and each chunk separately into one fixed vector; relevance is just the inner product of the two vectors. Chunk vectors can therefore be computed once, offline, and searched quickly, but the whole meaning of a chunk has to be squeezed into one vector before the model ever sees the question, so details such as "not", "unless", or a specific role get blurred.

A **Cross-Encoder** takes **[query] [SEP] [chunk]** as a single input, so self-attention runs across query and chunk tokens together. Each query word can attend to the exact chunk of words that answer it, which lets it judge relevance much more precisely. But nothing can be precomputed, it needs one full forward pass for every (query, chunk) pair.

2. Qualitative & rank-shift analysis

Rank changes (gold chunk = chunk containing the answer sentence; `results/rank_shifts.csv`)

Query (failure type)	**Gold chunk**	**Baseline rank → Re-ranked**	**In Top-3 before → after**
Q1 Who bears the cost of skilling electronics workers? (keyword trap: generic skilling / training chunks outrank it)	#81	**9 → 1**	No → Yes
Q2 Other ways to lodge complaints besides the app? (synonym / buried)	#169	**4 → 1**	No → Yes
Q3 Do tender relaxations apply to all bidders or only certain ones? (exception / scope)	#58	**4 → 1**	No → Yes
Q4 Is Telangana not planning to relax norms for SMEs? (negation)	#58	**4 → 1**	No → Yes

Query (failure type)	Gold chunk	Baseline rank → Re-ranked	In Top-3 before → after
Q6 Which departments must set aside budget for emerging tech? (buried)	#159	**9 → 1**	No → Yes
Q5 Which services still need a visit to an office? (negation)	#152 / #151	4 → 2 / 3 → 10	Yes → Yes
Q7 Who is not eligible for IT-park status? (role limit)	#63	6 → 5	No → No (**not fixed**)
Q8 Exception to the 60:40 ratio? (control)	#63	1 → 1	Yes → Yes
Q9 Is blockchain used in any welfare scheme? (control)	#213 / #212	1 → 2 / 2 → 6	Yes → Yes (gold ranked **lower**)

In 5 of 9 queries the critical chunk was outside the baseline Top-3 and the Cross-Encoder put it at rank 1. In Q1 the gold chunk was ranked 9th, behind chunks about re-skilling / emerging-tech training in general (ranks 1–2) and TASK's customised training programmes (rank 3), which share the words "skilling", "industry" and "workforce" but say nothing about who pays. The cross-encoder score for the gold chunk was the only positive one (+1.08 vs. about −2.5 to −11 for the rest).

Answer 1 (baseline) vs. Answer 2 (re-ranked) — `results/answers.md`

Query	Answer 1 (baseline Top-3)	Answer 2 (re-ranked Top-3)
Q1	Said the context "does not specify who bears the cost" — **omitted the key fact** (TASK trains on a no-cost basis).	Correct: TASK provides the workforce on a no-cost basis.
Q2	Named only the web application of Janahitha — **misread the question** (that is the app the user excluded) and **missed** telephone, SMS, WhatsApp, email, post, paper.	Correct and complete list of the other channels.
Q3	"Does not specify" — omitted the SME clause.	Partly correct: mentions the relaxation is for SMEs, but still hedges that scope is "not specified".
Q4	**Wrong:** "context does not mention any plans to relax norms for SMEs" — a	Correct: relaxation of experience, turnover, EMD, performance-guarantee norms.

Query	Answer 1 (baseline Top-3)	Answer 2 (re-ranked Top-3)
	false negation, because the relaxation clause was not in the context.	
Q5	Both give the same, correct answer: no service mandates presence *unless a physical test or inspection is required* (this exception is in chunk #11, page 2, which both pipelines retrieved).	Same. Re-ranking made no difference here.
Q6	"Does not specify which departments" — omitted the clause.	Correct: each user department allocates a special budget.
Q7	Individual landowners who do not wish to promote IT activity, plus applicants who do not submit Annexure 1 documents.	Same first point, then admits the context does not list other ineligible groups. Neither is clearly right (see caveats).
Q8, Q9	Correct.	Correct, essentially identical.

Summary: In these runs the baseline did not invent facts. Its errors were mostly **omissions**("the context does not say…", Q1, Q3, Q6), plus one **misstated** answer (Q2) and one **false negation** (Q4). All of these happened when the critical chunk was outside the Top-3, and each was fixed (Q3 only partly) once the Cross-Encoder promoted that chunk. Retrieval, not the LLM, was the bottleneck.

Caveats (honest limitations)
- **Q7 was not fixed**: The policy only lists who can apply; there is no explicit "not eligible" sentence, so even the Cross-Encoder ranked the right chunk 5th. This is a limit of the question/document, not of re-ranking.

- **Re-ranking can hurt rank (Q9, Q5):** In Q9 the Cross-Encoder pushed gold chunk #212 from 2 → 6 and a different chunk (#159) to rank 1. The answer stayed correct only because the duplicate-overlap chunk #213 stayed in the Top-3. In Q5 it demoted the heading-only chunk #151 (3 → 10) but kept #152, which has the same sentence.

- The `keyfact_*` columns in `answers.csv` are a crude keyword check and are wrong for Q4 (Answer 1 contains the word "relax" while saying the opposite) and Q5; the table above is based on reading the answers.

- Small corpus (307 chunks), 9 queries, one run, greedy decoding, gold chunks found by regex — these results are illustrative, not a statistical evaluation. The Q6 category label in the notebook says "rank 7", but the measured baseline rank was 9.

3. Latency profiling

NVIDIA A100-40GB, fp16 encoders, retrieval timings averaged over 20 runs per query after warm-up (`results/latency*.csv`); LLM generation is a single run per answer.

Stage	Mean	Median	Min – Max
Pipeline 1: baseline retrieval (query embedding + FAISS Top-10)	**7.1 ms**	7.1 ms	6.8 – 7.5 ms
Pipeline 2 extra step: Cross-Encoder on 10 pairs	**8.7 ms**	8.7 ms	8.3 – 9.0 ms
Pipeline 2 retrieval total (7.1 + 8.7)	**15.7 ms**	15.7 ms	15.3 – 16.5 ms
LLM generation, Answer 1	1.42 s	1.5 s	0.9 – 1.9 s
LLM generation, Answer 2	1.24 s	1.3 s	0.8 – 1.9 s

Re-ranking roughly doubles retrieval time (+8.7 ms, 2.2×), but that is under 1% of the end-to-end time, which is dominated by the LLM (about 1.2–1.4 s). The small difference between the two generation times comes from different answer lengths and is not a real effect of re-ranking.

4. Scaling question

A **Cross-Encoder** must read each (query, chunk) pair together, so no chunk representation can be computed in advance and the cost per query grows linearly with the corpus size N.

A **Bi-Encoder** embeds the documents once offline; at query time it needs only one query embedding plus a (sub-linear, with ANN indexes) vector search.

Measured here: scoring all 307 chunks with the **Cross-Encoder took 97 ms (0.31 ms/pair)**, against **15 ms for "Bi-Encoder Top-10 + Cross-Encoder on 10"**. That is already 6× slower for a 22-page document. **Linear extrapolation takes about 31 s per query for 100,000 chunks and about 0.9 hours for 10 million chunks**, on an A100 with batching (`results/scaling.csv`) — impossible for interactive use and very costly per query.

So the design is a funnel: the cheap Bi-Encoder gives high recall over everything, and the expensive, more accurate Cross-Encoder only re-ranks a small candidate set (here 10). The limitation is that the Cross-Encoder can only reorder what the first stage retrieved: a relevant chunk that is outside the Top-10 (e.g. chunk #80 in Q1, or #160 in Q6) can never be recovered by it.